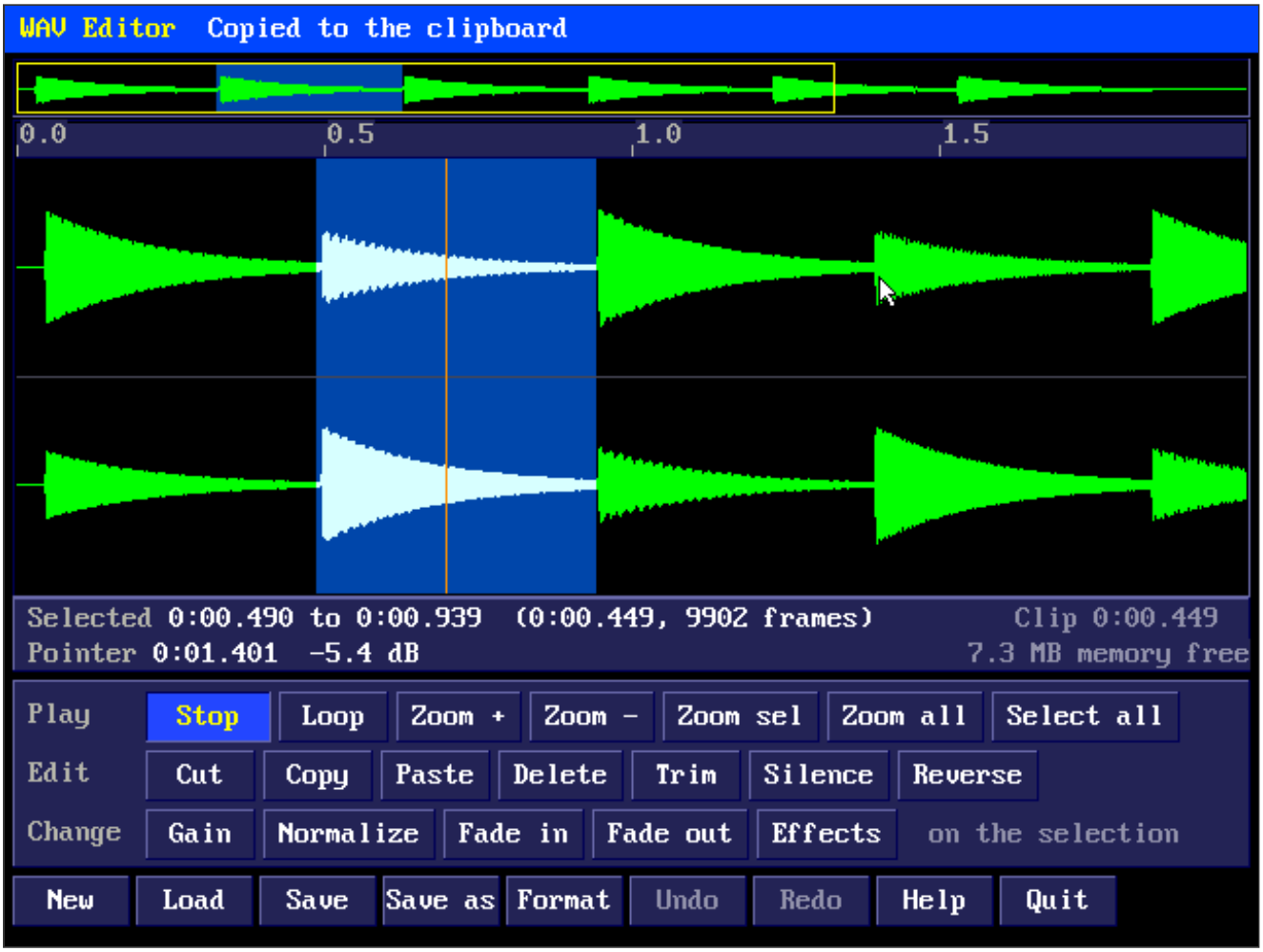
Pico 2 mini OS WAV Editor

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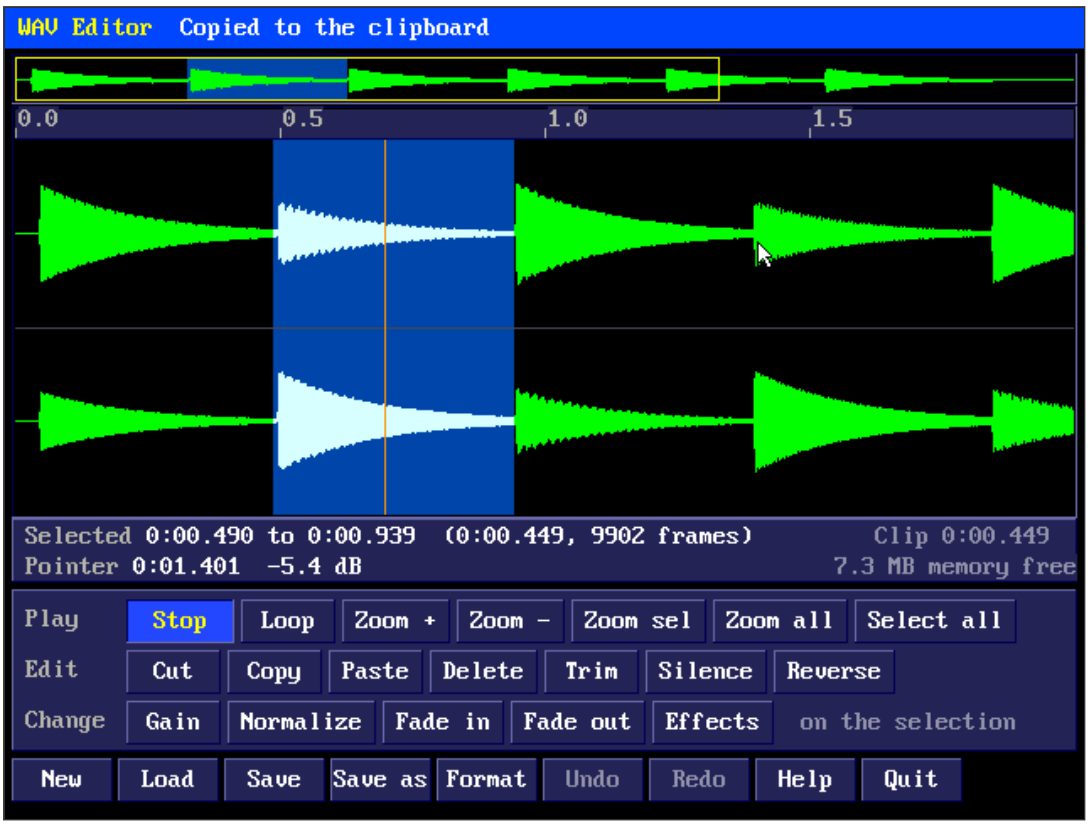
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A sound editor for WAV files, worked with the mouse: see the whole sound and zoom into it, select part of it, cut, copy, paste, delete and trim, change its level (gain, normalize, fades), add effects (echo, reverb, chorus, flanger, tremolo, filters, distortion), reverse it or make it silent, change its rate or make it mono or stereo, undo and redo, and hear it through the HDMI sound as you go.

The editor



`wav` (or `wavedit`) opens the editor on an empty sound (44100 Hz, mono, 16-bit); `wav DRUMS.WAV` opens that file (or starts a new one by that name). It needs a USB mouse; the keys do the rest.

From the top:

- **The title strip:** the file (with `*` when it has changes not saved), its rate, bits, mono or stereo, and its length; messages for a few seconds.
- **The overview:** the whole sound, with the selection shaded and the part shown in the waveform outlined in yellow. Click or drag on it to move the view there; the wheel over it scrolls.
- **The time ruler**, in seconds (minutes:seconds when zoomed out).
- **The waveform:** one lane, or two for stereo (left above, right below). Zoomed out, each column of the screen shows the lowest and highest sample under it; zoomed right in (16 screen pixels a frame), the line between the samples. The selection is shaded blue with the sound in it drawn white; the cursor (where a paste goes when nothing is selected) is a yellow line, and the play position an orange one.
- **Two lines** about the selection (its start, end, length in time and in frames) or the cursor, the clipboard's length, the time and level (dB below full scale) under the pointer, and the memory free.
- **The buttons:** Play (Stop while playing), Loop, Zoom + and -, Zoom sel (the selection fills the view), Zoom all and Select all; Cut, Copy, Paste, Delete, Trim, Silence and Reverse; Gain, Normalize, Fade in and Fade out, and Effects. Buttons that can't do anything now are greyed.

- **The file buttons** along the bottom, as in the sprite editor: New, Load, Save, Save as, Format, Undo, Redo, Help (the keys) and Quit.

The mouse

	Does
Left button on the waveform	puts the cursor there; drag to select (past either end, the view follows)
Double click	selects all of it
Right button	moves the nearer end of the selection there (drag to keep moving it)
Middle button	plays from there
Wheel on the waveform	zooms in and out around the pointer
Click or drag on the overview	moves the view there; the wheel there scrolls

Playing

Play (or Space) plays the selection, or with none selected from the cursor to the end; Space or **Stop** stops it, and so does Esc. With **Loop** on (L), the selection plays round and round (with none, the whole sound from the start each time). The view follows the play position a page at a time. The sound goes out through the HDMI sound at 48 kHz, whatever the file's rate (it is worked out between the samples), at half its level so that loud sounds don't clip on the way.

Editing

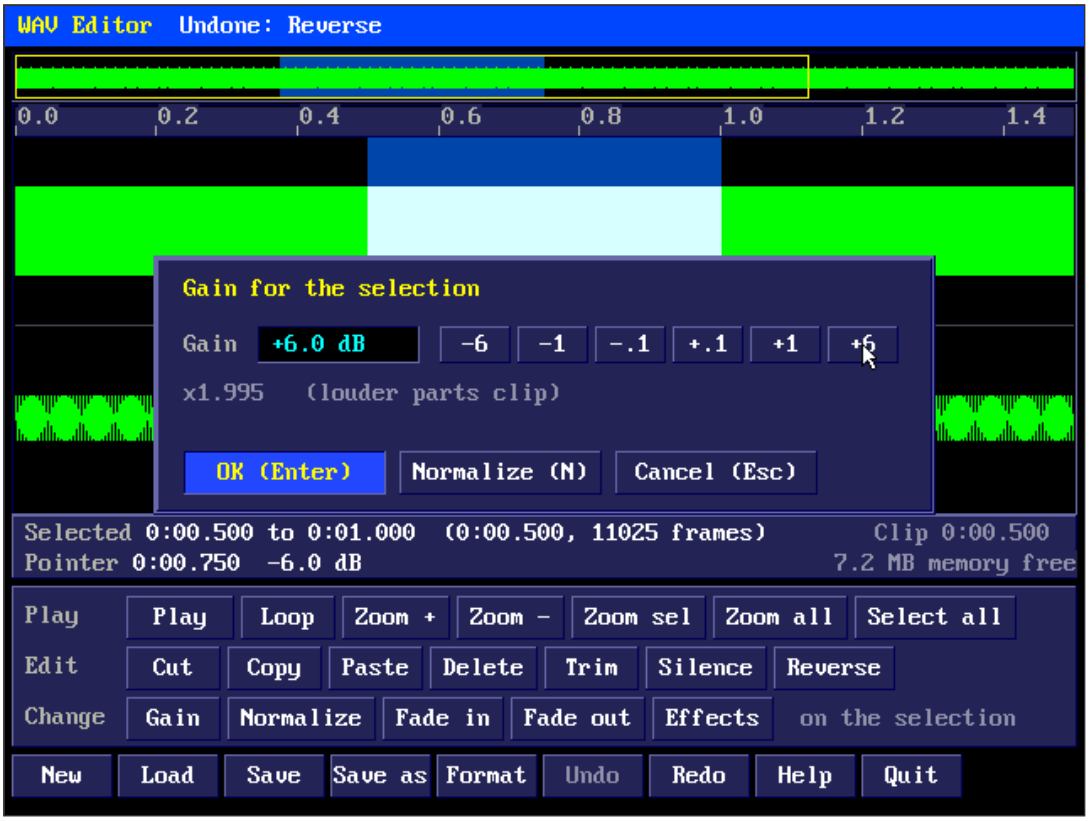
Button	Key	Does
Cut	Ctrl-X	the selection to the clipboard, and out of the sound
Copy	Ctrl-C	the selection to the clipboard
Paste	Ctrl-V	the clipboard in at the cursor, in place of the selection if there is one; the pasted part is left selected
Delete	Del, Backspace	the selection out of the sound
Trim	T	keeps only the selection
Silence	S	makes the selection silent; with none selected, puts a second of silence in at the cursor
Reverse	R	backwards

The clipboard stays while the editor runs, so it can go from one file to another (Load in between). Pasted mono sound becomes the same on both sides of a stereo one; stereo pasted into mono is mixed. The clipboard's samples are pasted as they are, so a clip from a file of another rate plays faster or slower there (change the rate of one first with Format).

Changing the level

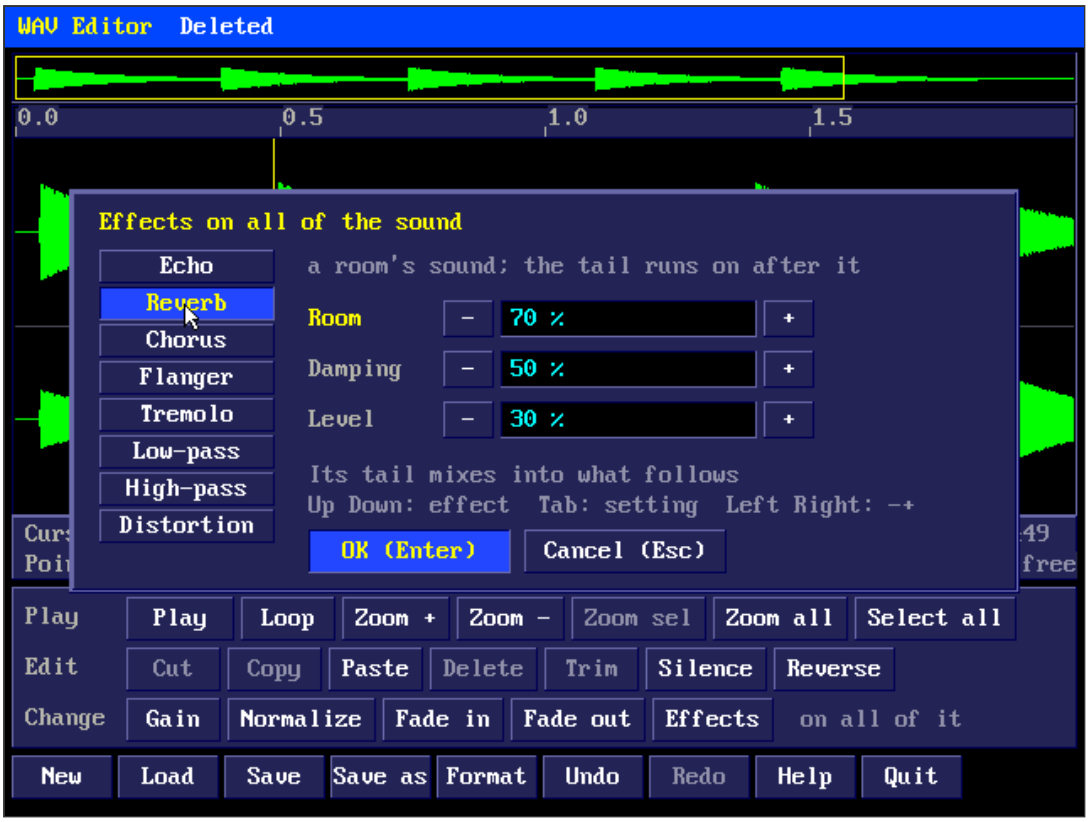
These work on the selection, or on all of the sound if nothing is selected (the panel says which).

Button	Key	Does
Gain	G	louder or quieter by the dB chosen in the gain box
Normalize	N	the loudest sample made full scale, the rest in step
Fade in	I	from silence at the start to as it was at the end, in a straight line
Fade out	O	from as it was to silence at the end



The gain box goes from -48 to +48 dB: -6, -1, -.1, +.1, +1 and +6 (or Left/Right a dB, Up/Down a tenth, PgUp/PgDn 6 dB, or the wheel), and shows what that multiplies by. +6 dB is about twice as loud, -6 dB half. Samples that would go past full scale are held at it (they clip), so Normalize is the way to make something as loud as it can be without clipping. **Normalize** in the box (N) normalizes instead.

Effects



Effects (E) opens a box with the effects down the left and the one chosen's settings on the right. Like the level changes, an effect works on the selection, or on all of the sound if nothing is selected. Click an effect (or Up/Down), set it with its - and + buttons (or Tab to choose a setting, Left/Right to change it, PgUp/PgDn ten steps, or the wheel), and OK (Enter) does it; Esc leaves the box. Each effect keeps its settings while the editor runs, so a change can be undone, tweaked and done again.

Effect	Settings	Does
Echo	Delay (10-2000 ms), Feedback (0-90 %), Level (0-100 %)	repeats of the sound a delay apart, each quieter by the feedback
Reverb	Room (0-100 %), Damping (0-100 %), Level (0-100 %)	the sound of a room (a Freeverb: eight combs and four all-passes a side); a bigger room rings longer, more damping makes it darker
Chorus	Depth (1-20 ms), Speed (0.1-5 Hz), Level (0-100 %)	a copy of the sound, delayed by a wavering 20 ms or so, mixed in: thicker and, in stereo, wider
Flanger	Depth (1-10 ms), Speed (0.1-5 Hz), Feedback (0-90 %)	a copy delayed by a sweeping few ms, mixed back: the jet-plane sweep
Tremolo	Speed (0.1-20 Hz), Depth (0-100 %)	the level going up and down
Low-pass	Cutoff (50-20000 Hz)	takes out the high notes above the cutoff (a 12 dB an octave filter)
High-pass	Cutoff (20-10000 Hz)	takes out the low notes below it

Effect	Settings	Does
Distortion	Drive (1-50), Level (0-100 %)	overdrive: the louder parts rounded off (a tanh curve), mixed with the sound as it was by the level

Echo, reverb, chorus and flanger go on after the sound ends: their tail (an echo's until its repeats are 60 dB down, a reverb's 0.5 to 3 seconds by the room, at most 10 seconds) is mixed into whatever follows the selection, and if it runs past the end of the sound, the sound gets that much longer. Samples that would go past full scale are held at it, so lower the level (or use Gain first) if an effect clips.

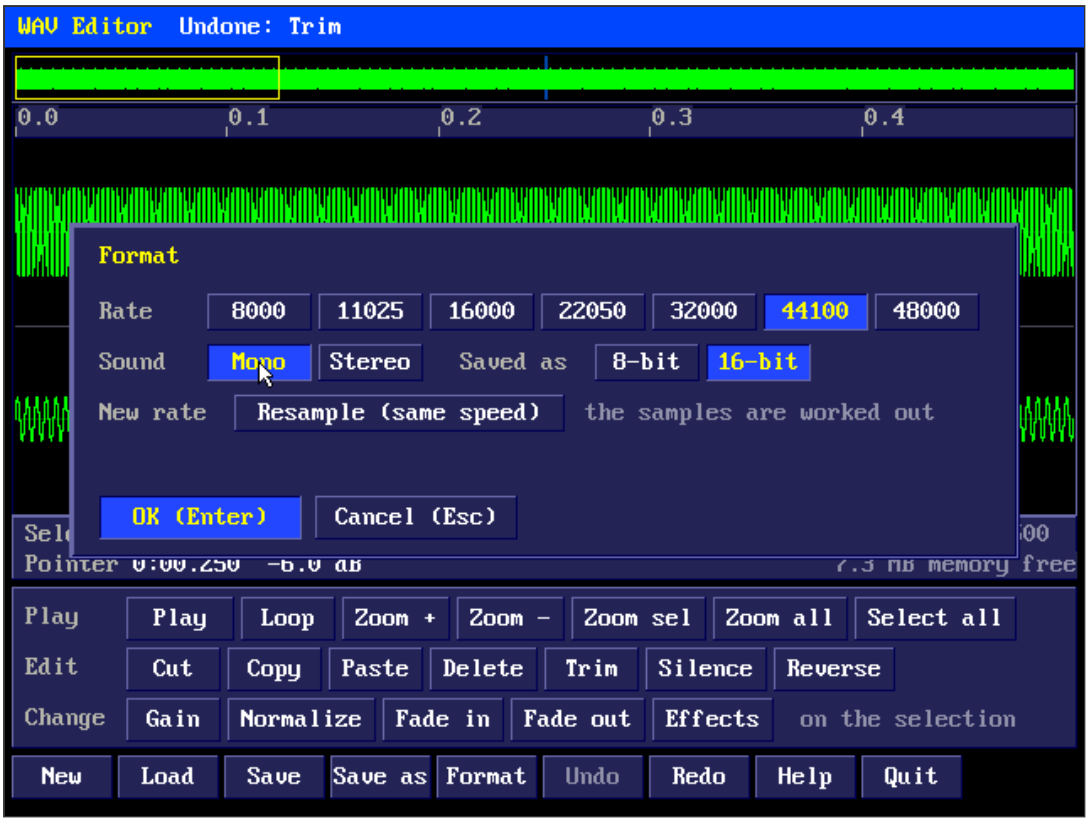
Undo

Undo (U or Ctrl-Z) and Redo (Ctrl-Y) go back and forward 32 steps: every edit, effect and format change. Each step keeps a copy of the sound it changed (a format change keeps the whole sound as it was), so undo uses memory: when the memory runs short, the oldest steps are forgotten to make room, and a change too big to keep a copy of is done all the same, saying "too big to keep for undo". Loading or New forgets them all.

The view

Key	Does
+ / - or Up / Down	zoom in / out (around the selection or cursor)
Z	zoom to the selection
A	zoom all (the whole sound)
Left / Right	move the view an eighth of the way
PgUp / PgDn	a page
Home / End	the cursor to the start / the end, and the view with it
Ctrl-A	select all

Format

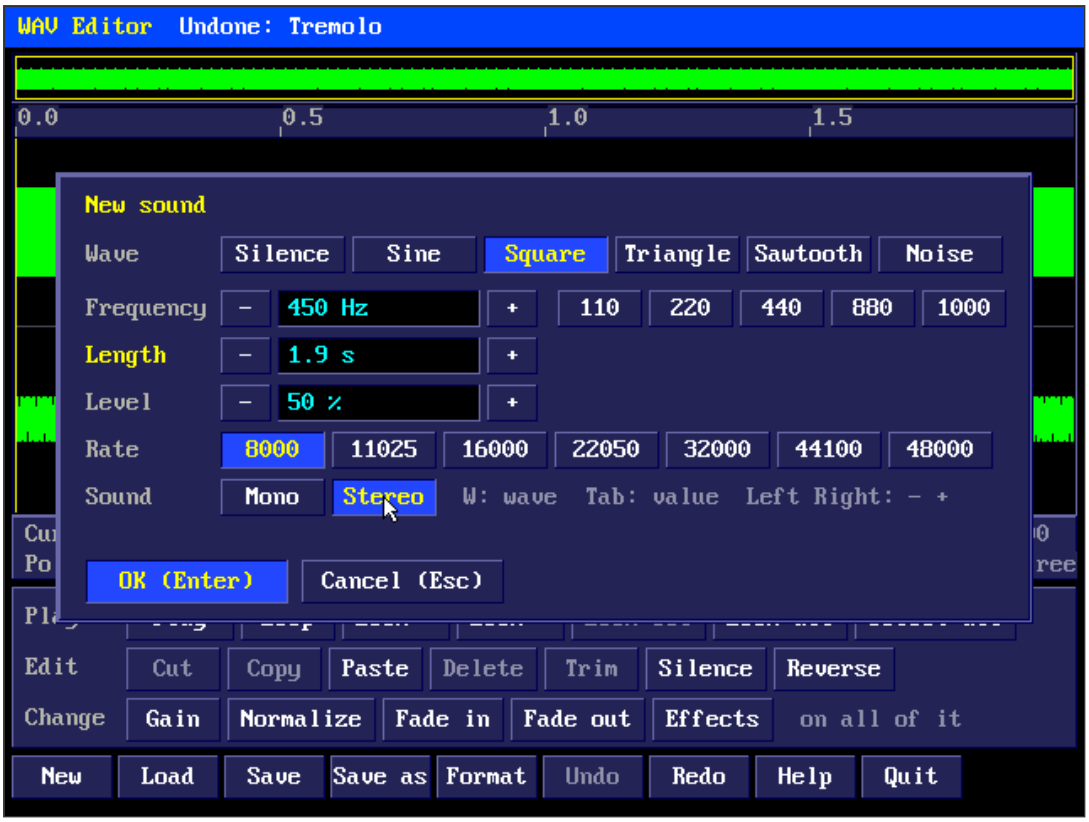


Format (F) changes the rate (8000, 11025, 16000, 22050, 32000, 44100 or 48000 Hz; Left/Right), mono or stereo (M, S), and whether Save writes 8 or 16-bit samples (8, 1). A new rate is either:

- **Resample (same speed)**: the sound is worked out again for the new rate, so it sounds the same (higher rates in a straight line between the samples; lower ones averaging the samples each new one covers, which keeps out most of the harshness of just dropping them); or
- **Relabel (speed changes)** (R): the samples are kept and only the rate changes, so the sound plays faster and higher (or slower and lower).

Stereo made mono mixes the two sides; mono made stereo puts the same on both. Format changes can be undone.

New

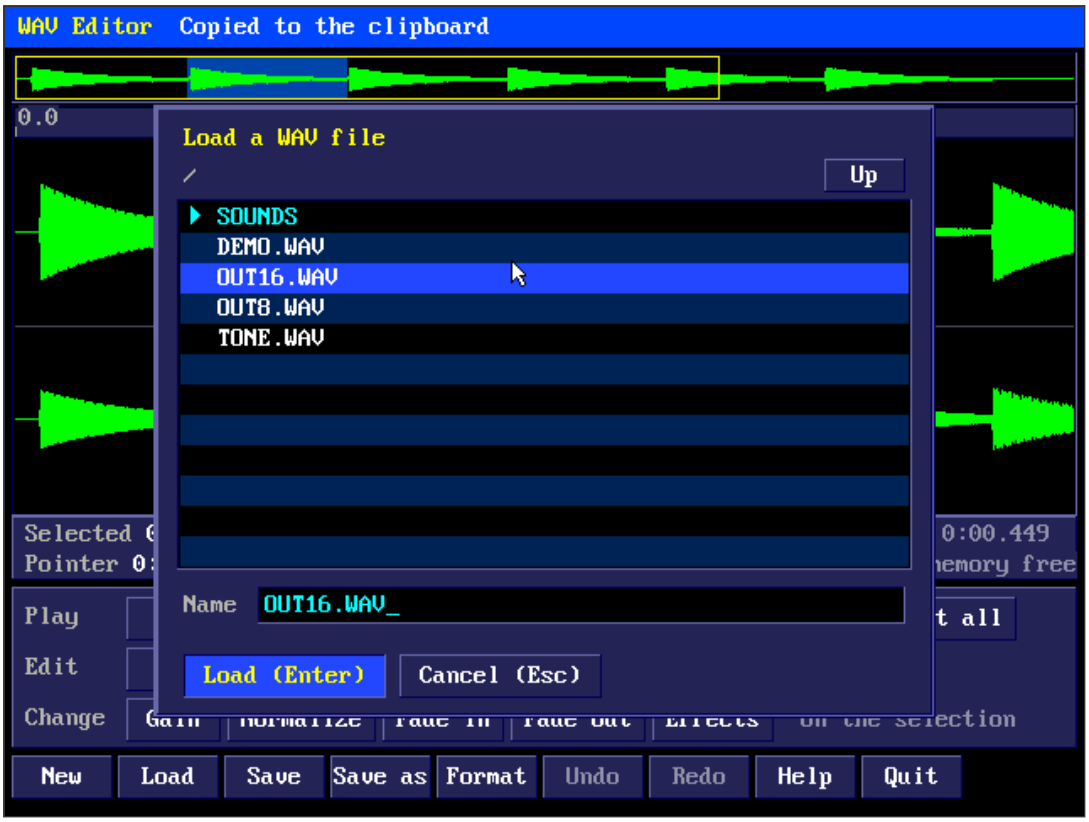


New (Ctrl-N) asks about changes not saved, then opens the New box, which starts the sound as a waveform:

- **Wave:** Silence, Sine, Square, Triangle, Sawtooth or Noise (W steps through them).
- **Frequency:** 1 to 20000 Hz, by 1 Hz below 100, 10 Hz below 1000 and 100 Hz above; or 110, 220, 440, 880 or 1000 at a click.
- **Length:** 0.1 s to 10 minutes (as long as the memory allows).
- **Level:** 1 to 100 % of full scale (50 % leaves room for effects).
- **Rate** and **Mono** or **Stereo** (M, S): the sound's format.

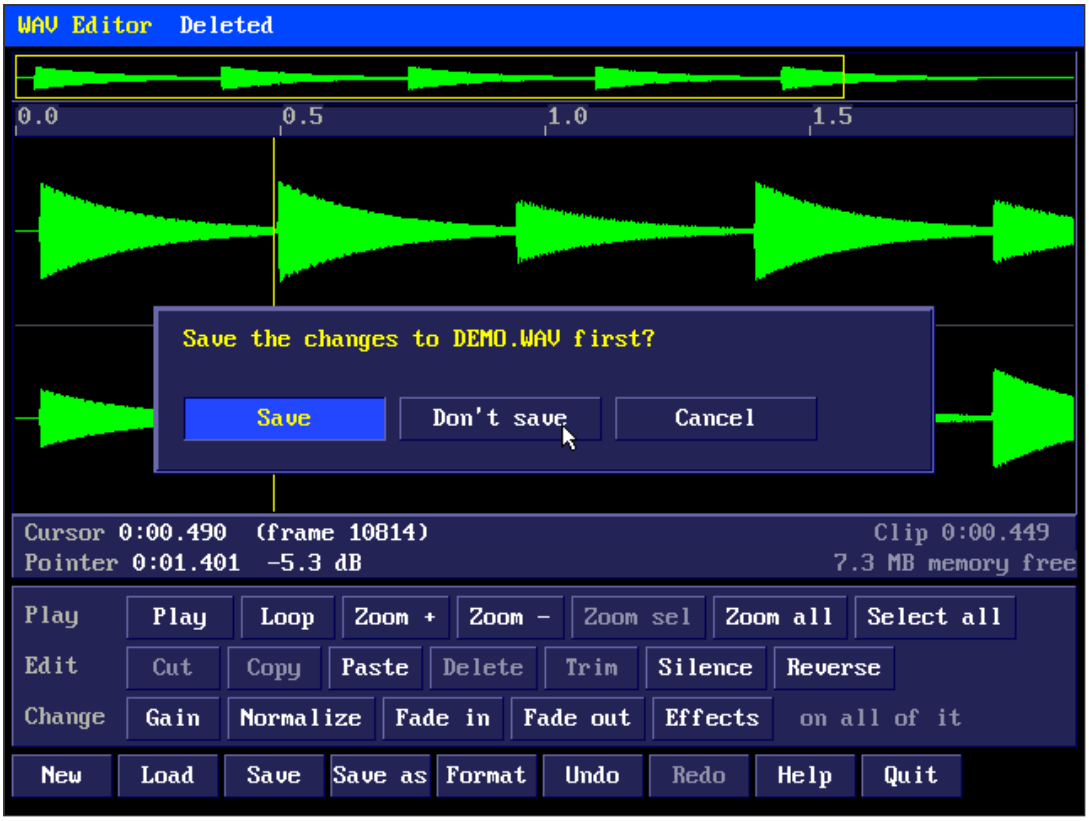
Tab (or Up/Down) chooses the value that Left/Right (or the wheel) change, PgUp/PgDn ten steps; Enter makes the sound, Esc keeps the one there. The box remembers its choices while the editor runs. Noise is white noise; the others start at the beginning of a cycle (the square and sawtooth with their jumps, as they are). Shape it with the effects and the fades: a sine through tremolo, a sawtooth through the low-pass, noise through the high-pass and a fade out for a hi-hat.

Loading and saving



Load and Save as open a file box on the file's folder (or the shell's): folders first, then the .WAV files. Click a folder to go into it, Up (or Backspace) to go back; click a file to choose it (twice to load it), or type a name. Enter loads or saves; a name without an extension gets .WAV. Saving over another file asks first.

Load reads PCM WAV files of 8, 16, 24 or 32 bits and 32-bit float, mono or stereo, at any rate from 1000 to 192000 Hz; of a file with more channels, the first two are kept. The editor keeps 16-bit samples, so 24-bit, 32-bit and float files are kept to 16 bits (the message says so). Save writes a plain PCM WAV: 16-bit, or 8-bit if the file was 8-bit or Format says so, at the sound's rate.



New, Load and Quit ask about changes not saved: **Save** saves them first (asking for a name if the sound has none), **Don't save** goes on without them.

Keys



Help (or F1) shows them in the editor too.

Key	Does
Space	play / stop
L	loop on or off
Ctrl-X, Ctrl-C, Ctrl-V	cut, copy, paste
Del or Backspace	delete the selection
T	trim to the selection
S	silence (with none selected: put in a second)
R	reverse
G	gain
N	normalize
I / O	fade in / fade out
E	effects
+ - Up Down, Z, A	zoom in / out, to the selection, all
Left Right PgUp PgDn	move the view
Home / End	the start / the end
Ctrl-A	select all
U or Ctrl-Z / Ctrl-Y	undo / redo
F	format
Ctrl-S or F2	save
F3 or Ctrl-O	load
Ctrl-N	new (the New box: a waveform, its frequency and length)
F1	the keys
Esc	stop playing; when not playing, quit (asks first if there are changes to save)

How long a sound can be

The sound is kept in PSRAM as 16-bit samples, so it can be as long as the memory free allows (the second line shows it; about 7 MB with nothing else running):

Sound	A second takes	7 MB holds about
8000 Hz mono	16 KB	7 minutes
22050 Hz mono	44 KB	2.7 minutes
22050 Hz stereo	88 KB	80 seconds
44100 Hz stereo	176 KB	40 seconds

Undo needs room too: an effect on all of a sound keeps a copy of all of it. With a long sound, select the part to change, or expect the oldest undo steps to be forgotten. A file too long to load says how much memory it needs.

From the shell

```
wav                         the editor, on an empty sound
wav DRUMS.WAV               the editor, on that file
wav info DRUMS.WAV          DRUMS.WAV: 22050 Hz, 16-bit, 2 channels, 0:02.000 (44100
frames)
```

How it works

The editor keeps the sound, the clipboard, the undo steps, a summary of the sound (the lowest and highest sample of every 512 frames, for the overview and for drawing it zoomed out quickly) and a picture of the whole screen in PSRAM while it runs, and nothing at all once it ends.

This board's PSRAM has had a few cells that forget after a minute or more untouched. So, as the text editor does with its text, every 8 KB of the sound, the clipboard and the undo steps in use has a checksum (the room kept past the end of the sound, never written, isn't checked) (kept in the Pico's own memory while the editor runs), and every 15 seconds each block is read from the chip, checked and written back, which renews it. If a block has changed by itself, the title strip turns dark red and says **MEMORY ERROR**, and Save won't write over the file it came from: Save as a new file, and compare.

Playing runs on core 1 in the sound mixer, at 48 kHz: each output sample is worked out in a straight line between the two samples of the sound around it. The waveform's columns are worked out when the view changes; dragging a selection redraws only the waveform, straight onto the screen, and the play position moves by redrawing the two columns it leaves and enters.

`tools/wavtest/uitest.sh` builds the editor on a PC and works it with a simulated mouse and keys, checking loading, selecting, cut, copy, paste, delete, trim, the New box's waveforms, every level change and effect (an echo's repeats exactly, a reverb's tail, the filters' cutoffs) and their undo and redo, format changes, saving 8 and 16-bit files and reading them back, playing, the checksums catching a changed bit, and running out of memory (under ASan); it writes the pictures in this manual.